CSE4062
Introduction to Data Science and Analytics
Spring 2023

DELIVERY 2

IT Support Tickets

MEMBERS

1. Data Preprocessing

The data provided contains some missing and noisy values. We applied following steps to clean the data.

- Firstly, converted Excel file to CSV file. And parsed it by using Pandas.
- Compilation time column used to have negative or NaN values, replaced them with mean of the compilation time column (only positive values are used to calculate the mean)
- Missing values are filled with the next valid value of column. It filled missing values only if they have valid values after them.
- Missing values are filled with the last valid value of column. If filled missing values only if they have valid values before them.
- By using Pandas' *drop_duplicates* function we removed duplicate rows from the dataset.
- The new dataset is saved to a new CSV file for using it in the future steps.

2. Feature Selection

There are 18 columns in the dataset including target column *URGENCY*. We picked them 5 columns to use in our training process by using different methods. Before running feature selection algorithms we did one-hot encoding to categorical values of the dataset. Then we determined the columns will be used in the training by using random forests, information gain and manual selection. Here is the list of selected features.

Table 1. Selected features

	Random Forests (I)	**Information Gain (II)**	**Manual Selection (III)**
IMPACT_Operation Continue With Workaround	x	x	
IMPACT_Some Processes Stop on Production	x		x
PRIORITY_Major	x	x	x
PRIORITY_Minor		x	

WAITING_HOURS_AT_CUSTOMER	x	x	
Completion Time	x	x	x
ISSUE_TYPE_Incident			x
COMMENT_COUNT			x

3. Models

- Support Vector Machine
- Stochastic Gradient Descent
- Decision Tree
- Naive Bayes
- Random Forest
- Multi-Layer Perceptron

4. Results

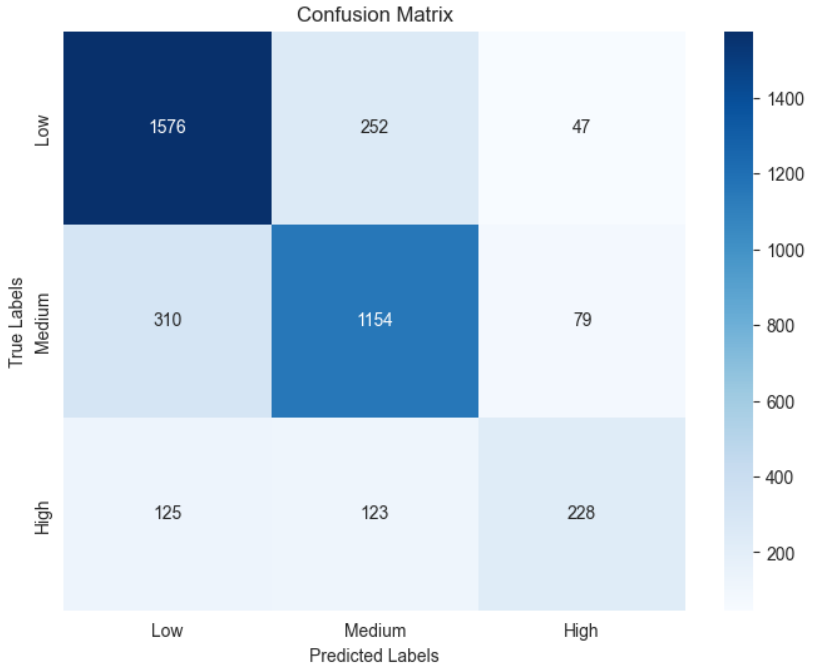
Table 2. Performances of each algorithm tested with three different feature subset sorted by F1 scores

Algorithm	Features	Accuracy	F1	Precision	Recall	F1-Micro	F1-Macro
Random Forest	I	0,76	0,75	0,75	0,76	0,76	0,70
Random Forest	II	0,72	0,71	0,71	0,72	0,71	0,66
Decision Tree	I	0,70	0,70	0,70	0,70	0,71	0,65
Decision Tree	II	0,68	0,67	0,67	0,68	0,68	0,62
MLP	I	0,68	0,67	0,67	0,68	0,70	0,62
Random Forest	III	0,65	0,65	0,64	0,65	0,65	0,57
Decision Tree	III	0,65	0,64	0,64	0,65	0,64	0,57

MLP	III	0,64	0,63	0,64	0,64	0,64	0,53
MLP	II	0,66	0,62	0,65	0,66	0,66	0,59
SGD	III	0,59	0,55	0,52	0,59	0,58	0,41
Naive Bayes	II	0,60	0,54	0,57	0,60	0,60	0,41
SGD	I	0,57	0,54	0,55	0,57	0,54	0,43
SGD	II	0,56	0,52	0,54	0,56	0,59	0,43
Naive Bayes	III	0,51	0,51	0,44	0,46	0,51	0,34
SVM	III	0,53	0,47	0,47	0,53	0,53	0,35
Naive Bayes	I	0,54	0,47	0,51	0,54	0,54	0,37
SVM	I	0,51	0,41	0,58	0,51	0,51	0,30
SVM	II	0,51	0,41	0,58	0,51	0,51	0,30

The best model is random forest with the feature set I. The confusion matrix of it's the best fold can be seen below.

Figure 1. Confusion matrix of the best model



The second best model (with a different algorithm) is decision tree with the feature set I. The t_test scores between them given below.

Table 3. T-test between random forest model and decision tree model

t-statistic	10.597172769757407
p-value	7.638353991567266e-19

Figure 2. All metrics of all models when feature subset I is used

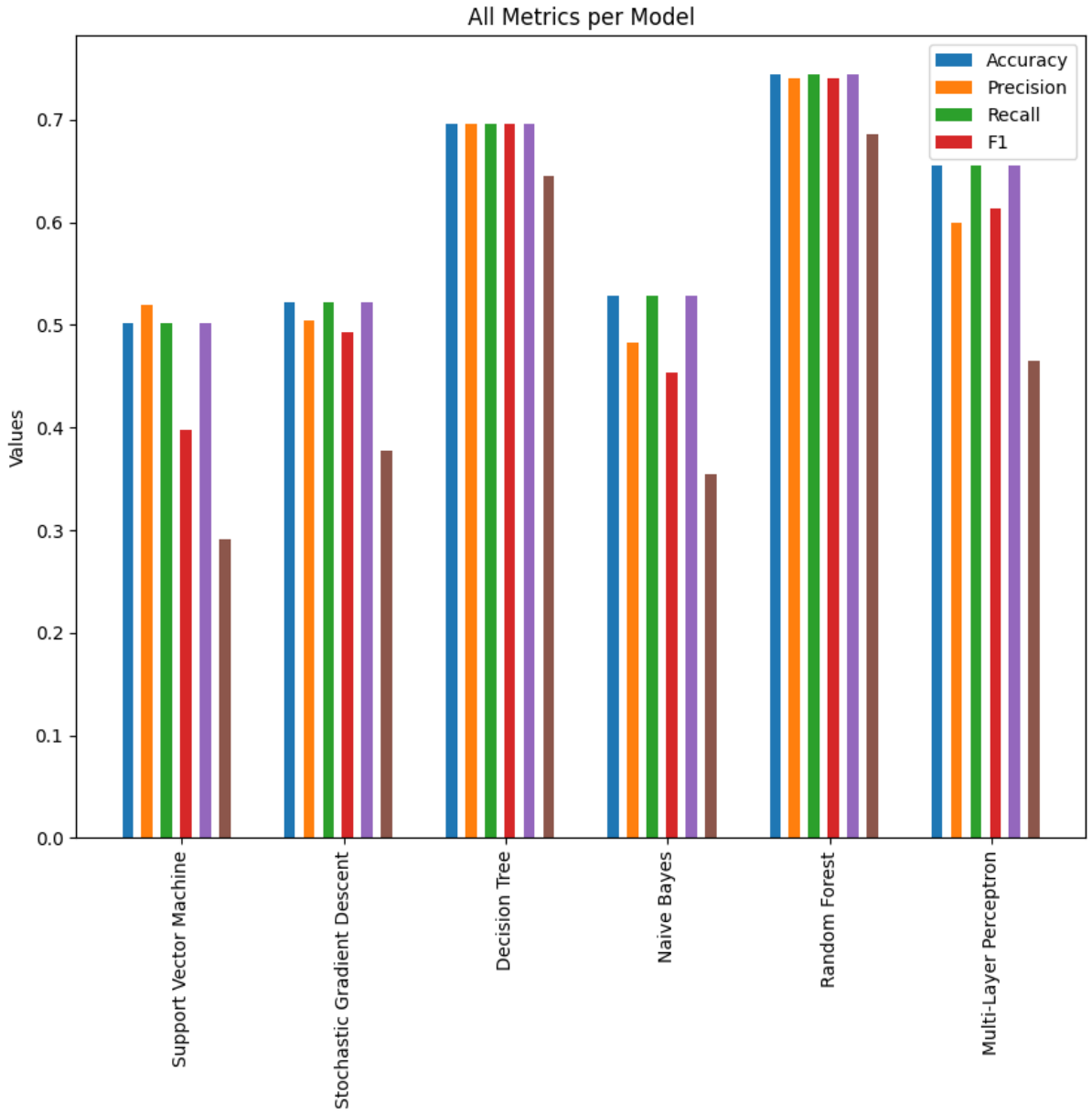
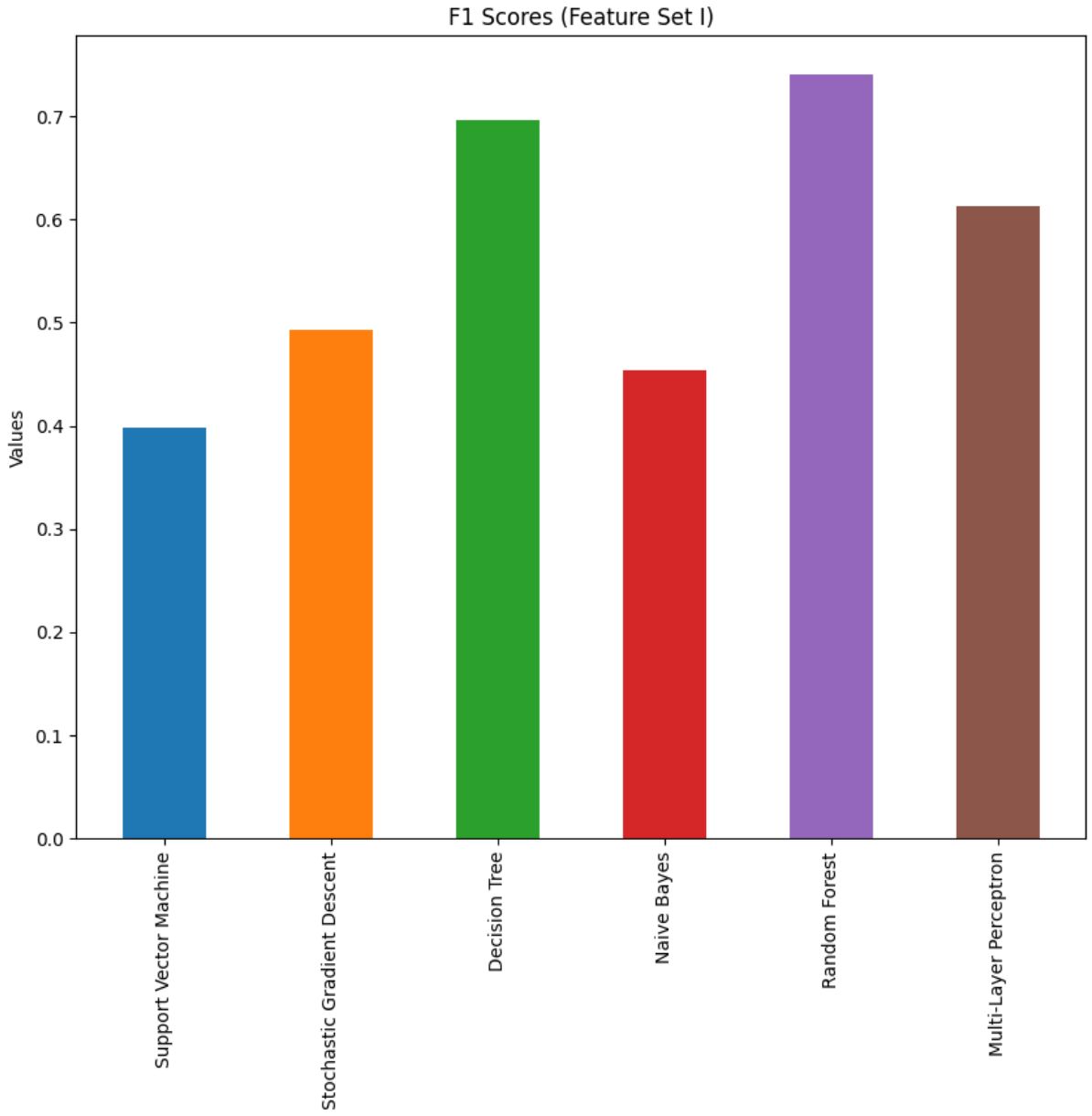


Figure 3. F1 scores of different models when feature subset 1 is used



5. Conclusion

For each feature subset random forest model was the best followed by decision tree model. In average feature subset I has 0.591 F1 score followed by subset II with 0,581 F1 score. The third subset has 0,576 F1 score. We can say that random forests was the best feature selection algorithm for this dataset.

Random Forest (I) achieved the highest accuracy of 0.76, making it the top-performing algorithm in the list.

Random Forest (II) follows closely with an accuracy of 0.72, indicating strong performance but slightly lower than the first Random Forest variant.
Decision Tree (I) achieved an accuracy of 0.70, making it the third-highest performing algorithm in the list.

The t-statistic represents the difference between the means of the two results (random forest and decision tree) relative to the variability within each group. A larger t-statistic indicates a larger difference between the means. In this case, the t-statistic is quite large, suggesting a significant difference between the two models.

The p-value is a measure of the statistical significance of the t-statistic. It indicates the probability of observing such extreme results (or more extreme) if there were no true difference between the means. A smaller p-value suggests stronger evidence against the null hypothesis (no difference between the means). In this case, the p-value is extremely small (close to zero), indicating strong evidence to reject the null hypothesis and suggesting a significant difference between the two models.

Therefore, based on these results, we can conclude that there is a statistically significant difference between the random forest model and the decision tree model and the random forest model is the best model.